

Analyse IV - Summary

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1 Information

A concise and rigorous summary transcribing the most important concepts and theorems from the Analyse IV lecture at EPFL.

2 Holomorphic and Analytic Functions

2.1 Definitions

Let $\Omega \subseteq \mathbb{C}$ be an open set. A complex function $f : \Omega \rightarrow \mathbb{C}$ is said to be **complex differentiable** at a point $z_0 \in \Omega$ if the following limit exists:

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

If f is complex differentiable at *every* point of Ω , we say f is **holomorphic** on Ω . If f is holomorphic on the entire complex plane ($\Omega = \mathbb{C}$), we call f an **entire function** (e.g., polynomials, e^z , $\sin(z)$).

3 The Cauchy-Riemann Equations

Let $z = x + iy \in \mathbb{C}$. A complex function $f(z)$ can be separated into its real and imaginary parts:

$$f(x + iy) = u(x, y) + iv(x, y)$$

where $u, v : \mathbb{R}^2 \rightarrow \mathbb{R}$.

3.1 Necessary and Sufficient Conditions

Theorem: Suppose $u(x, y)$ and $v(x, y)$ have continuous first-order partial derivatives (i.e., they are of class C^1) on an open set Ω . The function $f = u + iv$ is holomorphic on Ω if and only if it satisfies the **Cauchy-Riemann equations**:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

When these equations hold, the complex derivative can be computed as:

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

3.2 The Complex Chain Rule

The chain rule for complex functions works exactly like the chain rule in real calculus. This is a vital tool for proving that a composite function is holomorphic without having to check the Cauchy-Riemann equations directly.

The Formula:

Let $g : \Omega \rightarrow U$ be holomorphic on $\Omega \subseteq \mathbb{C}$, and let $f : U \rightarrow \mathbb{C}$ be holomorphic on U . Then the composite function $h(z) = (f \circ g)(z) = f(g(z))$ is holomorphic on Ω , and its derivative is:

$$h'(z) = f'(g(z)) \cdot g'(z)$$

3.2.1 How to Use the Chain Rule

To compute the derivative of a complex composition, follow these steps:

1. **Identify the components:** Break the function into an "outer" function $f(w)$ and an "inner" function $g(z)$.
2. **Differentiate the outer function:** Compute $f'(w)$ with respect to its variable w .
3. **Differentiate the inner function:** Compute $g'(z)$ with respect to z .
4. **Substitute and Multiply:** Multiply the results together, replacing the variable w in $f'(w)$ with the original inner function $g(z)$.

Example: Differentiating $h(z) = e^{z^2}$

- Let the outer function be $f(w) = e^w \implies f'(w) = e^w$.
- Let the inner function be $g(z) = z^2 \implies g'(z) = 2z$.
- Applying the formula:

$$h'(z) = e^{g(z)} \cdot (2z) = 2ze^{z^2}$$

3.2.2 Example: Logarithmic Composition

Consider $f(z) = \log(1 + z^2)$ using the principal branch of the logarithm.

1. Domain and Holomorphicity:

The principal branch $\log(w)$ is holomorphic for $w \in \mathbb{C} \setminus (-\infty, 0]$. Thus, $f(z)$ is holomorphic where $1 + z^2$ is not a non-positive real number. This occurs when $z \neq iy$ for $|y| \geq 1$. The largest open subset of holomorphicity is:

$$\Omega = \mathbb{C} \setminus \{iy : y \in \mathbb{R}, |y| \geq 1\}$$

2. Derivative via Chain Rule:

Let $w = g(z) = 1 + z^2$. Then $f(z) = \log(w)$.

$$\begin{aligned} f'(z) &= \frac{d}{dw}(\log w) \cdot \frac{d}{dz}(1 + z^2) \\ &= \frac{1}{1 + z^2} \cdot 2z = \frac{2z}{1 + z^2} \end{aligned}$$

Note on Holomorphicity:

If g is holomorphic on Ω and f is holomorphic on $g(\Omega)$, then $f \circ g$ is guaranteed to be holomorphic on Ω . This allows us to conclude that functions like $\sin(e^z)$ or $\cos(z^3)$ are **entire functions** because they are compositions of entire functions.

3.3 Harmonic Conjugates

Because holomorphic functions are infinitely differentiable, u and v have continuous second-order derivatives. By applying the Cauchy-Riemann equations and Schwarz's theorem (symmetry of second derivatives), we get:

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{and} \quad \Delta v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Therefore, the real and imaginary parts of a holomorphic function are **harmonic functions**. We say that v is the harmonic conjugate of u .

4 The Complex Logarithm

4.1 Principal Argument

For any $z \neq 0$, the principal argument $\text{Arg}(z)$ is the unique angle in the interval $(-\pi, \pi]$. It is computed using the arctangent function depending on the quadrant of (x, y) .

Crucial limitation: $\text{Arg}(z)$ is discontinuous across the negative real axis.
this is how we compute it :

$$\text{Arg}(z) = \begin{cases} \arctan\left(\frac{y}{x}\right) & \text{if } x > 0 \\ \arctan\left(\frac{y}{x}\right) + \pi & \text{if } x < 0 \text{ and } y \geq 0 \\ \arctan\left(\frac{y}{x}\right) - \pi & \text{if } x < 0 \text{ and } y < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \text{ and } y > 0 \\ -\frac{\pi}{2} & \text{if } x = 0 \text{ and } y < 0 \end{cases}$$

4.2 Principal Branch of the Logarithm

We define the **principal branch** of the complex logarithm, denoted $\text{Log}(z)$, as:

$$\text{Log}(z) = \ln |z| + i\text{Arg}(z) = \ln\left(\sqrt{x^2 + y^2}\right) + i\text{Arg}(x + iy)$$

Rigorous Properties:

- **Domain:** $\text{Log}(z)$ is a well-defined, single-valued function on the cut plane $\mathbb{C} \setminus (-\infty, 0]$. We must remove zero and the negative real axis (the branch cut) to maintain continuity.
- **Holomorphy:** On this cut plane, $\text{Log}(z)$ is holomorphic.
- **Derivative:** Its complex derivative is exactly what we expect from real analysis:

$$\frac{d}{dz}\text{Log}(z) = \frac{1}{z} \quad \text{for } z \in \mathbb{C} \setminus (-\infty, 0]$$

5 Lecture 3

6 Complex Integration

Definition (Line Integral)

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a regular (smooth) curve that parametrizes the path $\Gamma \subseteq \mathbb{C}$. Let $f : \Gamma \rightarrow \mathbb{C}$ be a continuous function. We define the line integral of f along Γ by:

$$\int_{\Gamma} f(z) dz = \int_a^b f(\gamma(t))\gamma'(t) dt$$

Example

Let Γ be the unit semi-circle centered at $z = 0$, parametrized counterclockwise: $\gamma(t) = e^{it}$ for $t \in [0, \pi]$. Then $dz = ie^{it}dt$.

6.1 Cauchy's Theorem

Let $U \subseteq \mathbb{C}$ be an open and simply connected set, and let γ be a closed and piecewise regular curve inside U . If $f : U \rightarrow \mathbb{C}$ is holomorphic, then:

$$\oint_{\Gamma} f(z) dz = 0$$

- **Simply connected:** The set consists of "one piece" and has no holes.
- **Piecewise regular:** Γ consists of finitely many smooth curves joined end-to-end (e.g., the boundary of a square).

6.1.1 Example

Let Γ be the unit circle and $f(z) = z^2$. Since f is holomorphic on the simply connected set $U = \mathbb{C}$, by Cauchy's Theorem:

$$\oint_{\Gamma} z^2 dz = 0$$

6.2 The Extension of Cauchy's Theorem (Deformation)

Let Γ_0, Γ_1 be closed, simple, regular curves such that $\Gamma_1 \subseteq \text{int}(\Gamma_0)$. Let γ_0, γ_1 be parametrizations of Γ_0, Γ_1 with the same orientation. If f is holomorphic on the region between Γ_0 and Γ_1 (including the boundaries), then:

$$\oint_{\Gamma_0} f(z) dz = \oint_{\Gamma_1} f(z) dz$$

6.3 Cauchy's Integral Formula

Suppose $D \subseteq \mathbb{C}$ is open and $f : D \rightarrow \mathbb{C}$ is holomorphic. If $z_0 \in D$, then:

$$f(z_0) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - z_0} dz$$

for every simple, closed, counterclockwise-oriented curve γ in D whose interior contains z_0 .

7 Lecture 4

8 Tricks to Compute Integrals Easily

- **Cauchy's Theorem:** If f is holomorphic on a **simply connected** set D , then for any closed loop γ in D :

$$\int_{\gamma} f(z) dz = 0$$

- **Deformation Theorem:** We can replace an integral over a "complicated" curve Γ_0 by an integral over a "simpler" curve Γ_1 (like a small circle) as long as f is holomorphic in the region between Γ_0 and Γ_1 .

- **Path Independence:** If Γ_0 and Γ_1 share the same endpoints and orientation, and f is holomorphic on the region they enclose, then:

$$\int_{\Gamma_0} f(z) dz = \int_{\Gamma_1} f(z) dz$$

- **Generalized Cauchy Integral Formula:** Useful for computing integrals of the form $\frac{f(z)}{(z-z_0)^{n+1}}$. If f is holomorphic on and inside γ :

$$\int_{\gamma} \frac{f(z)}{(z-z_0)^{n+1}} dz = \frac{2\pi i}{n!} f^{(n)}(z_0)$$

Example: To calculate $\int_{\gamma} \frac{e^z}{z^3} dz$ around 0, use $n = 2$ and $f(z) = e^z$. The result is $\frac{2\pi i}{2!} f''(0) = \pi i$.

9 Taylor Series and Laurent Series

9.1 Taylor Series

Real case: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be $N+1$ times differentiable. For any $x_0 \in \mathbb{R}$, the Taylor polynomial is $T_{x_0}^N f(x)$. If $f \in C^{N+1}$, there exists $\theta \in (0, 1)$ such that:

$$f(x) = \sum_{n=0}^N \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n + \frac{f^{(N+1)}(\theta x_0 + (1 - \theta)x)}{(N+1)!} (x - x_0)^{N+1}$$

Complex case: Let $f : D \rightarrow \mathbb{C}$ be holomorphic. If the disk $B_R(z_0)$ is contained in D , then f is analytic and:

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n \quad \text{for all } z \in B_R(z_0)$$

The largest such R is the **Radius of Convergence**.

9.2 Laurent Series

Theorem: Let f be holomorphic on an annulus (punctured disk) $0 < |z - z_0| < R$. Then:

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n = \dots + \frac{c_{-2}}{(z - z_0)^2} + \frac{c_{-1}}{z - z_0} + c_0 + c_1(z - z_0) + \dots$$

where the coefficients are given by:

$$c_n = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z_0)^{n+1}} dw$$

Definition: The series is split into two parts:

- **Singular part:** Terms with negative powers ($n < 0$).
- **Regular part:** Terms with non-negative powers ($n \geq 0$).

10 Classification of Singularities and Residues

10.1 Types of Isolated Singularities

By looking at the singular part of the Laurent series of f at z_0 :

1. **Removable Singularity:** No negative powers ($c_n = 0$ for all $n < 0$).
2. **Pole of order k :** The lowest power is $c_{-k} \neq 0$ (finitely many negative terms).
3. **Essential Singularity:** Infinitely many negative terms (e.g., $e^{1/z}$ at $z = 0$).

10.2 The Residue Theorem

The coefficient c_{-1} is called the **Residue** of f at z_0 , denoted $\text{Res}(f, z_0)$.

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{k=1}^m \text{Res}(f, z_k)$$

This is the most powerful "trick" to solve integrals where γ encloses several poles z_k .